



Probabilistic Models for Some Intelligence and Attainment Tests

GEORG RASCH

**WITH A FOREWORD AND
AFTERWORD BY
BENJAMIN D. WRIGHT**

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FOREWORD

Benjamin D. Wright

Probabilistic Models for Some Intelligence and Attainment Tests is the most important work on psychometrics since Thurstone's articles of 1925-1929 and the 1929 monograph by Thurstone and Chave. The psychometric research done by Rasch between 1951 and 1959, which he explains and illustrates in this book, marks the point at which psychometrics moved from being purely descriptive to become a science of objective measurement. The psychometric grail of psychological measures that transcend the questions or items producing them was glimpsed by Thurstone in the 1920s when he set down his requirements for valid measuring.

A measuring instrument must not be seriously affected in its measuring function by the object of measurement. To the extent that its measuring function is so affected, the validity of the instrument is impaired or limited. If a yardstick measured differently because of the fact that it was a rug, a picture, or a piece of paper that was being measured, then to that extent the trustworthiness of that yardstick as a measuring device would be impaired. Within the range of objects for which the measuring instrument is intended, its function must be independent of the object of measurement.

(Thurstone 1928a, page 547; 1959, page 228)

Objective measurement, that is, measurement that transcends the measuring instrument, not only requires measuring instruments which can function independently of the objects measured, but also a response model for calibrating their functioning, which can separate instrument and object effects. But, because Thurstone and his followers did not parameterize or condition for persons individually in their response models, they never attained objectivity. Even today there are psychometricians who try to achieve objective measurement with methods built on samples of normally distributed random individuals and response models that deny parameter separation. But they fail to grasp deductions (Rasch 1968, 1977; Andersen 1973a, 1977; Barndorff-Nielsen 1978) which show the necessity and sufficiency for objectivity of linear logistic response models with no interaction terms. It was the Danish mathematician, Georg Rasch, who first understood the possibilities for objectivity that reside in the logistic response model with one item parameter and one person parameter, and it was Rasch who first applied this model to the analysis of mental test data.

Georg Rasch, Doctor of Philosophy in mathematics (1930), member of the International Statistical Institute (1941), charter member of the Biometric Society (1947), professor of statistics (1962), and knight of the Order of Dannebrog (1967), was born in Odense, Denmark, on 21 September 1901, the youngest and, according to his father and himself, the "least practical" of three brothers.* His mother was ill when he was

* I am extremely grateful to my colleague and friend David Andrich for these quotations. They come from an interview with Rasch which Andrich recorded in Laesoe in June 1979, supplemented by a letter Rasch wrote to me 5 February 1980.

young, and Rasch has few recollections of her influence on his childhood. His fiercely religious father, however, left a deep impression. Wilhelm Rasch, sailor, ship's officer, mathematics teacher and self-appointed missionary, was, according to his son, "one of the most hard-boiled evangelists I have ever known."

Wilhelm Rasch moved his family to Svendborg in 1906 to open a mission high school for prospective seamen. It was there in 1914 that Georg became fascinated by his father's forgotten trigonometry texts and was fortunate to find a teacher who made arithmetic and algebra, "which could so easily be extremely boring something with which a wonderful world was opened."

The geometry behind the trigonometry did not interest Georg. But the algebra in the trigonometric manipulations fascinated him. Fortunately his arithmetic teacher realized that he was a born mathematician and persuaded his father, despite the extra expense, to send Georg to the cathedral school in Odense where there was a curriculum specializing in mathematics and science. He enrolled there in 1916 and in 1919 graduated and went to the University of Copenhagen.

In Copenhagen I entered the Faculty of Science, to which mathematics belonged, and got into immediate contact with my teachers. I had, of course, to learn the elements of function theory and even geometry, but I concentrated upon the analytic part which I liked. My first professor lectured on functions, algebra, and number series. . . . Although I worked for him on his numerical studies, what caught my interest was the theory of Lagrange equations. That resulted in my first publication, a joint paper by my professor and myself (Nielsen and Rasch 1923). . . .

I got a stipend for my further studies and became a member of college Regensen, where we received free room and board. When I got the stipend I did not see any further reason for doing arithmetical work for my earnings, so I left Professor Nielsen and got another teacher, Professor Nørlund, who had written an extremely good book on difference equations. Nørlund was my professor for the rest of my time as a student, and I was his assistant teacher from 1925, when I graduated, until 1940. The topics in function theory that Nørlund lectured about together with the other topics I had to study in order to lecture as his assistant built up my mathematical background.

Nørlund was also the director of the Geodetic Institute . . . to which I became attached to provide mathematical and computational assistance. This added to my income and in February 1928, I married my sweetheart, Elna Nielsen, with the charming call-name of Nille. Two daughters were added to the family in 1931 and 1933.

My thesis, which I defended in 1930, was the fruit of my cooperation with Nørlund, but in a field which he himself did not cultivate at the time. It dealt with matrix algebra and its applications to the theory of linear systems of difference and differential equations. I have always loved to think, but I have never been inclined to do extensive reading and so I had never seen anything about

matrices. Nørlund gave lectures on systems of difference equations in which he wrote out every equation in detail every time. When working through my notes I discovered, to my surprise, that these long equations could be condensed in a very simple way. I did not know anything about matrices at that time, but just invented them for myself and discovered what their rules must be. Only later did I find out that others had already formalized the idea.

So I invented my own theory of matrices, especially as they applied to linear systems of difference and differential equations of first order. The . . . part of my thesis on the theory and application of product integrals which developed the solution of a linear system of differential equations as a generalization of the ordinary elementary integral was published in German (Rasch 1934). Many years later I learned that the techniques developed in this paper played a part in solving some problems in atomic theory and were also used to prove some difficult theorems in group theory. I mention this to point out that, although I have been known as a statistician, my original talent, if I had any, must have been in mathematics.

The early 1930s were difficult for Rasch. Aside from teaching as Nørlund's assistant and some small matters in the seismic division of the Geodetic Institute, there was no work in mathematics. Rasch had, however, two medical acquaintances studying the reabsorption of cerebrospinal fluid in monkeys. Helping them to understand their data gave him his first practical experience with the exponential distribution and material for his first experimental paper (Fog, Rasch, and Stürup 1934).

The success of this collaboration encouraged Fog and Stürup to engage Rasch to teach a small group of psychiatrists and neurologists some elementary mathematics and statistics. Word of this got to the head of the Hygienic Institute, who was also interested in statistics. The outcome was that Rasch served the Hygienic Institute as statistical consultant from 1934 to 1948 and also became attached to the State Serum Institute, a relationship which continued until 1956.

About the same time Nørlund, for whom Rasch still taught mathematics, and Madsen, the director of the Serum Institute got into a conversation about him at a meeting. They agreed that, in order to do his job at the Serum Institute, he needed to learn the latest developments in statistics. One of them knew about R. A. Fisher, so they applied to the Rockefeller Foundation for a year's study with Fisher in London.

The Rockefeller fellowship was granted, but, while it was brewing, Rasch went to Oslo for three months on a Carlsberg grant to study Ragnar Frisch's confluence analysis, a technique developed for economics, but similar to Thurstone's factor analysis. Then, in September 1934, Rasch joined Fisher's staff at the Galton Laboratory in London.

I went through Fisher's statistical methods and learned the kind of chi squares . . . he used. . . . I soon got hold of his 1922 paper where he developed his theory of maximum likelihood, because I was especially interested in that matter. What caught my interest most was his idea that this is a form of generalization of just the same kind as Gauss attempted when he invented the method of least squares.

The meaning of least squares is not, in Fisher's interpretation, just a minimization of a sum of squares. It is a maximization of the probability of the observations, choosing such values as estimates of the parameters as will maximize the probability of the set of observations at your disposal. There is a very essential difference between this and the simple idea of minimizing sums of squares. This philosophy went further when Fisher got to his concept of sufficiency. . . .

To purely mathematical minds sufficiency may appeal as nothing more than a surprising and singularly nice property, extremely handy when accessible, but, if not, then you just do without it. But to me sufficiency means much more than that. When a sufficient estimate exists, it extracts every bit of knowledge about a specified feature of the situation made available by the data as formalized by the chosen model. 'Sufficient' stands for 'exhaustive' as regards the feature in question.

What is left over when a sufficient estimate has been extracted from the data is independent of the trait in question and may therefore be used for a control of the model that does not depend on how the actual estimates happen to reproduce the original data. This is a cornerstone of the probabilistic models that generate specific objectivity.

The realization of the concept of sufficiency, I think, is a substantial contribution to the theory of knowledge and the high mark of what Fisher did. . . . His formalization of sufficiency nails down the . . . conditions that a model must fulfill in order for it to yield an objective basis for inference.

During his year in London, Rasch also discussed the problem of relative growth with Julian Huxley. Using data on crab shell structure, Rasch had discovered that it was possible to measure the growth of individual crabs as well as populations. His conversations with Huxley convinced him that this was the line of research he wished to follow. "It meant quite a lot to me to realize the meaning and importance of dealing with individuals and not with demography. Later I realized that test psychologists were not dealing with the testing of individuals, but . . . were studying how traits, such as intelligence, were distributed in populations. They were making demographic studies and not studies of individuals."

Rasch began teaching statistics to biologists in the fall of 1936, but not until 1938 did he come into professional contact with psychologists. Another friend, Rasmussen, worked at the Psychological Laboratory at the University of Copenhagen, where there happened to be a six month vacancy. When the director, Professor Rubin, heard of Rasch's interest in statistics, he asked him to give a few lectures to the psychologists. This connection lasted for thirty years.

The war interfered with further developments, but in 1944 a program in educational psychology was begun at the laboratory and Rasch was engaged to teach statistics. This aroused his interest in the multivariate normal distribution and led him to an elegant proof of Wishart's theorem (Rasch 1948).

Meanwhile, a friendship with Chester Bliss made in London in 1935 brought Rasch to the United States in 1947 to participate in the founding of the Biometrics Society

(Rasch 1947a) and in the postwar reorganization of the International Statistical Institute in Washington. In the course of these meetings Tjalling Koopmans, a fellow student of Ragnar Frisch's confluence analysis and Fisher's sufficient statistics, invited Rasch to spend two months with the Coles Commission on Economic Research at the University of Chicago, where he met L. J. Savage. This set the stage for Savage to bring Rasch back to Chicago in 1960, to finish writing *Probabilistic Models* and to give the series of lectures that introduced this writer to Rasch's new psychometrics.

Rasch began his work on psychological measurement in 1945 when he helped Rubin and Rasmussen standardize an intelligence test for the Danish Department of Defense (Rasch 1947b). But,

the big event came in 1951 when the Minister of Social Affairs wanted to know whether the kind of extra education given to poor readers had any effect that endured. Children who had been to school in the 1940s were now around twenty years old. They were retested for reading capability a number of times. By comparing scores I could draw some conclusions about the development of each individual student. But that work was very imperfect. One of my collaborators collected a set of data from 125 children, where each had been followed for five years. Each year they were tested one or two times for their reading ability. I thought there should be some kind of developmental study in that. When I got the data I made the guess that it might be a good idea to try the multiplicative Poisson model. It turned out that it fitted the data quite well.

The next essential event was in 1953, when Borge Prien constructed an intelligence test consisting of four parts. Each of the parts were constructed to be closely conformable to the multiplicative model for dichotomous items. The discovery of that model occurred in connection with my analysis of the reading tests. I had chosen the multiplicative Poisson for the reading tests because it seemed a good idea mathematically, if it would work. It turned out that it did. Then I wanted to have some motivation for using it, and not only the excuse that it worked. In order to do so, I imitated the proof of a theorem concerning a large number of dichotomous events, each of them having a small probability. Under conditions which can be specified easily, including that these probabilities be small, the number of events becomes Poisson distributed. I imitated that proof, but in doing so I took care that the imitation ended up with the multiplicative Poisson model, that is, I made sure that there was a personal factor entering into each of the small probabilities for the dichotomous outcome. The probabilities for the dichotomous case should therefore be of the form $\lambda/(1 + \lambda)$, and the λ would have a factor that was personal through all of the items and each item, of course, would have its own parameter and then I had my new model.

I had taken a great interest in intelligence tests while working with Rubin and Rasmussen in 1945. It struck me that I might analyze the test we had constructed then, and which had been taken over by the Military Psychology Group. I tried

to see whether the model worked there and also whether it worked in some other tests. The first thing I did . . . was to analyze the Raven tests. They worked almost perfectly according to the multiplicative model for dichotomous items. That was my first nice example using the newly discovered model. Now I compared the results of the Raven's test and the results of my analysis of the military intelligence test. The intelligence test did not conform.

When I showed this to the head of the military psychologists he immediately saw the point. I had talked to him about my attempts to make sense of intelligence tests by means of the model I had discovered in connection with the multiplicative Poisson. And I had also told him about the Raven's tests. But now I presented the examination of the test he actually had in current use from the Psychology Laboratory. I pointed out to him that it would seem to consist of different groups of items with quite different kinds of subject matter. His immediate reaction was to call on Borge Prien who was working for the military psychologists and to give him the order that, within the next six months, before the next testing session in November 1953, to have ready a new intelligence test consisting of four different subtests, each of these to be built in such a way that they followed the requirements that Rasch demanded.

It was remarkable. Prien did that in six months. He invented tests, which, when you see them, are rather surprising. He really did invent items of the same sort, from very easy to very difficult, and spaced in a sensible way. We did do some checking in the process and omitted or modified items that did not seem to be working. It was really a masterpiece. Prien had been told, 'All you have to construct is four different kinds of tests, with very different subject matters and each of them should be just as good as Georg tells us that Raven's tests are.' And he did so. That was when I really began to believe in the applicability of that elementary model.

In 1957 I gave . . . some free lectures on the researches I had done since Prien's construction of the new intelligence tests. I told about the multiplicative Poisson and about the nice little model which sorts items out from each other. My lectures were tape-recorded, and my daughter Lotte got the task of deciphering them and writing them down. She made a proper work out of it, and what she did was taken over by the Educational Institute, and they had it mimeographed.

At that time the institute consisted of five different departments, each with its own head. Every Friday morning the company of them, together with the director, Erik Thomsen, and I had a meeting where we discussed current matters. Thomsen organized it so that on a number of these Fridays we went through my manuscript. That was very good for it. It clarified many points that I had been vague about. I was forced by the young fellows there to make it quite clear what I really meant.

By 1953 Rasch had used a Poisson model to analyze a family of oral reading tests and with Borge Prien had designed and built a four-test intelligence battery fitting

the requirements of his logistic model for item analysis. Rasch discussed his concern about sample dependent estimates in an article on simultaneous factor analysis in several populations (Rasch 1953). However, his work on item analysis remained unknown outside Denmark until 1960, when he lectured in Chicago, gave a paper at the Berkeley Symposium on Mathematical Statistics (Rasch 1961), and published *Probabilistic Models*.

In her 1965 review of person and population as psychometric concepts, Jane Loevinger wrote,

Rasch (1960) has devised a truly new approach to psychometric problems. . . . He makes use of none of the classical psychometrics, but rather applies algebra anew to a probabilistic model. The probability that a person will answer an item correctly is assumed to be the product of an ability parameter pertaining only to the person and a difficulty parameter pertaining only to the item. Beyond specifying one person as the standard of ability or one item as the standard of difficulty, the ability assigned to an individual is independent of that of other members of the group and of the particular items with which he is tested; similarly for the item difficulty. . . . Indeed, these two properties were once suggested as criteria for absolute scaling (Loevinger, 1947); at that time proposed schemes for absolute scaling had not been shown to satisfy the criteria, nor does Guttman scaling do so. Thus, Rasch must be credited with an outstanding contribution to one of the two central psychometric problems, the achievement of nonarbitrary measures. Rasch is concerned with a different and more rigorous kind of generalization than Cronbach, Rajaratnam, and Gleser. When his model fits, the results are independent of the sample of persons and of the particular items within some broad limits. Within these limits, generality is, one might say, complete.

(Loevinger 1965, page 151).

In Rasch's words:

In the beginning of the 60's I introduced a new—or rather a more definite version of an old—epistemological concept. I preserved the name of *objectivity* for it, but since the meaning of that word has undergone many changes since its Hellenic origin and is still, in everyday speech as well as in scientific discourse, used with many different contents, I added a restricting predicate: *specific*.

Let it be said at once: my professional background is mathematical and statistical, not philosophical. The concept has therefore not been carved out in a conceptual analysis, but on the contrary its necessity has appeared in my practical activity as a statistical consultant for about 30 years and in the later years as a professor of Theoretical Statistics with Reference to its Applications to the Social Sciences.

During these activities I was introduced to very diverse subjects: medicine and hygiene with the connected parts of biology; psychology and education; technology; economics; demography and sociology; linguistics; etc. In spite of the diversity of subjects the analytic methods generally available were rather

limited for the first many years. But in 1951 I was faced with a task the solution of which added a new tool to my arsenal.

In that year The Danish Ministry of Social Affairs wanted an investigation of the development of reading ability in 125 then about 20 years old former students of public schools in Copenhagen, who in their school years had suffered from serious reading difficulties and therefore had received supplementary education in that discipline.

For each of these students were recorded the results of repeated oral reading tests during his school years—both as regards reading speed and reading accuracy—as well as of a test at an after-examination late in 1951.

It would be a fairly simple task to follow the development of a student's reading ability over a number of years if the same part of the same test were used every time, but at each testing it was necessary to choose a test which corresponded approximately to his standpoint, so as a rule a student was followed up with a series of tests of increasing "degrees of difficulty".

In a concrete formulation of this problem I imagined—in good statistical tradition—the possibility that the reading ability of a student at each stage, and in each of the two above-mentioned dimensions, could be characterized in a quantitative way—not through a more or less arbitrary grading scale, but by a positive real number defined as regularly as the measurement of a length.

Whether this would be possible with the tests in question could not be known in advance. It had to be tried out through a separate experiment which was carried out in January 1952. In this experiment ab. 500 students in the 3rd–7th school year were tested with 2 or 3 of the texts used in the investigation of students with reading difficulties.

(Rasch 1977, pages 58–59)

Rasch develops his seminal analysis of these reading data in chapters I through IV of *Probabilistic Models*. Not only is his discussion easy to follow and to learn from, but it allows one to participate in the implementation of a decisive step in psychometric method. These four chapters describe the dawn of a new era in psychometric theory and practice.

The outcome of the reading test experiment was beyond expectation: a statistically very satisfactory analysis on the basis of a new model which represented a genuine innovation in statistical techniques!

But the understanding of what the model entails tarried several years. At the 1959 anniversary of the University of Copenhagen the highly esteemed Norwegian economist Ragnar Frisch—later Nobel Prize winner—was to receive an honorary doctorate. I visited him by appointment the next day, and when our business was finished he asked me what I had been doing in the 25 years since I stayed at his institute in Oslo for a couple of months to study a new technique of statistical analysis that he had developed.

As mentioned I had been doing some very varied things, but I soon concentrated on the comparison of reading abilities, one of the topics of a monograph

which I was then preparing.

On this occasion I did not mention reading errors, but rather the students' reading speeds. The model is, however, the same: the Multiplicative Poisson Model which I then proceeded to explain to Ragnar Frisch in the following way:

Applied to reading speed the model states that the probability that person no. v in a given time reads a_{vi} words of text no. i is determined by the Poisson distribution

$$p\{a_{vi}\} = e^{-\xi_v \cdot \epsilon_i} \cdot \frac{\xi_v^{a_{vi}} \cdot \epsilon_i^{a_{vi}}}{a_{vi}!} \quad (\text{II:1})$$

where a high value of the parameter $\lambda_{vi} = \xi_v \cdot \epsilon_i$ means that many words are read in the given time, and thus a high value of ξ_v means *high reading speed* and a large value of ϵ_i means a text which is quickly read, thus in this respect an *easy reading test*.

Likewise the probability that the person reads a_{vj} words of test no. j under similar conditions is

$$p\{a_{vj}\} = e^{-\xi_v \cdot \epsilon_j} \cdot \frac{\xi_v^{a_{vj}} \cdot \epsilon_j^{a_{vj}}}{a_{vj}!} \quad (\text{II:2})$$

It is also assumed that the actual results of test no. i , a_{vi} , has no influence on the probabilities of the possible outcome of the other test. The multiplication rule for probabilities in these conditions says that the probability of the outcomes a_{vi} and a_{vj} of the two tests is the product of the two probabilities (II:1) and (II:2) thus

$$p\{a_{vi}, a_{vj}\} = p\{a_{vi}\} p\{a_{vj}\} = e^{-\xi_v (\epsilon_i + \epsilon_j)} \cdot \frac{\xi_v^{a_{vi} + a_{vj}} \cdot \epsilon_i^{a_{vi}} \cdot \epsilon_j^{a_{vj}}}{a_{vi}! a_{vj}!} \quad (\text{II:3})$$

The Poisson distribution has the property (which in fact can be derived from (II:3)) that the sum of the two Poisson distributed variables is also Poisson distributed with a parameter which is the sum of the two parameter values. With the notation a_{v+} for the sum $a_{vi} + a_{vj}$, we then have

$$p\{a_{v+}\} = e^{-\xi_v (\epsilon_i + \epsilon_j)} \cdot \frac{\xi_v^{a_{v+}} \cdot (\epsilon_i + \epsilon_j)^{a_{v+}}}{a_{v+}!} \quad (\text{II:4})$$

At this stage we apply another basic rule of the calculus of probabilities which in this context can be formulated thus: In the class of possible outcomes where the total number of words read, a_{v+} , has a fixed value, the probability of the outcomes a_{vi} and a_{vj} , conditional on this total, is given by dividing (II:4) into (II:3).

Until now the non-mathematical reader has been advised to skip the formulas (and until this point Frisch had only listened politely), but now I shall present a crucial point which demands a careful inspection of the two last formulas, but not

necessarily an understanding of their content:

On the right side of (II:3) and (II:4) the person parameter ξ_r appears in exactly the same [way in the] two expressions, namely in the exponential function of the same argument:

$$e^{-\xi_r(\epsilon_i + \epsilon_j)} \quad (\text{II:5})$$

and raised to the same power:

$$\xi_r^{a_{r+}} \quad (\text{II:6})$$

When (II:4) is divided into (II:3) these two factors both cancel out, and the resulting conditional probability does not contain the person parameter ξ_r : the probability that the given number of words read, a_{r+} , is composed of a_{ri} and a_{rj} words of the two tests is expressed by

$$p\{a_{ri}, a_{rj} | a_{r+}\} = \left(\frac{a_{r+}}{a_{ri}}\right) \left(\frac{\epsilon_i}{\epsilon_i + \epsilon_j}\right)^{a_{ri}} \left(\frac{\epsilon_j}{\epsilon_i + \epsilon_j}\right)^{a_{rj}} \quad (\text{II:7})$$

which is determined by the observed numbers a_{ri} and a_{rj} and by the ratio between the difficulty parameters of the two tests, while it is not influenced by which person involved. On seeing (II:7) Frisch opened his eyes widely and exclaimed: "It (the person parameter) was eliminated, that is most interesting!" And this he repeated several times during our further conversation. To which I of course agreed every time—while I continued reporting the main results of the investigation and some of my other work.

Only some days later I all of a sudden realized what in my exposition had caused this reaction from Ragnar Frisch. And immediately I saw the importance of finding an answer to the following question: "Which class of probability models has the property in common with the Multiplicative Poisson Model, that one set of parameters can be eliminated by means of conditional probabilities while attention is concentrated on the other set, and vice versa?"

What Frisch's astonishment had done was to point out to me that the possibility of separating two sets of parameters must be a fundamental property of a very important class of models.

(Rasch 1977, pages 63–66)

It is the two earliest and most popular members of this "very important class of models" which Rasch presents and applies in *Probabilistic Models*. Although the book focuses on the measurement of reading accuracy, speed, and intelligence, the basic principles employed are fundamental to all scientific work.

When first suggesting the models (for measuring) I could offer no better excuse for them than their apparent suitability, which showed in their rather striking mathematical properties. In Rasch (1961) a more general point of view was indicated, according to which the models were strongly connected with what seemed

to be basic demands for a much needed generalization of the concept of measurement.

In continuation of that paper my attention was drawn to other fields of knowledge, such as economics, sociology, history, linguistics, evaluation of arts, etc. where claims are arising of being taken just as seriously as Natural Sciences.

On a first sight the observational material in Humanities would seem very different from that in physics, chemistry and biology, not to speak of mathematics. But it might turn out that the difference is less essential than it would seem. In fact, the question is not whether the observations are of very different types, but whether Sciences could be firmly established on the basis of quite different types of observation.

(Rasch 1967)

The psychometric methods introduced in this book go far beyond measurement in education or psychology. They embody the essential principles of measurement itself, the principles on which objectivity and reproducibility, indeed all scientific knowledge, are based.

PREFACE

For several years statistical methods have been a favourite instrument within various branches of psychology. Warnings have, however, not always been wanting. Two instances from recent literature may serve as examples.

Skinner¹ vigorously attacks the application of statistics in psychological research, maintaining that the order to be found in human and animal behavior should be extracted from investigations into individuals, and that psychometric methods are inadequate for such purposes since they deal with groups of individuals.

As far as abnormal psychology is concerned Zubin² expresses a similar view in stating: "Recourse must be had to individual statistics, treating each patient as a separate universe. Unfortunately, present day statistical methods are entirely group-centered so that there is a real need for developing individual-centered statistics."

Individual-centered statistical techniques require models in which each individual is characterized separately and from which, given adequate data, the individual parameters can be estimated. It is further essential that comparisons between individuals become independent of which particular instruments – tests or items or other stimuli – within the class considered have been used. Symmetrically, it ought to be possible to compare stimuli belonging to the same class – "measuring the same thing" – independent of which particular individuals within a class considered were instrumental for the comparison.

This is a huge challenge, but once the problem has been formulated it does seem possible to meet it. The present work demonstrates, by way of three examples from test psychology, certain possibilities for building up models meeting these demands. And it would seem quite possible to modify and extend the methods used here to cover much larger areas, but in order to investigate how far the principles go – and what should be done outside possible limits – much research is needed. It is hoped, however, that planned continuations of the

present work and contributions from others will gradually enlarge the field where fruitful models can be established.

In part A, chapters I–VII, I have tried to discuss the basic principles rather carefully, using only the amount of mathematics that I deemed necessary – mathematical models can hardly be adequately presented quite without mathematics. In the same chapters the reader will find a rather detailed numerical and graphical documentation which may, I hope, serve as a guide for others in analyzing their own suitable data from similar fields. I should be very grateful for information about such studies, carried out or being planned.

In part B, chapters VIII–X, a fairly full account is given of the mathematical background, extended, however, for the sake of completeness, to cover some more ground than has actually been utilized in part A.

Psychologists wishing to use the methods in practice will miss a careful discussion of how to do so. I consider, however, that it is as yet a little early to go right ahead to practice. For one thing, the work presented here needs corroboration from other sources where different, but related test batteries have been applied to other groups of people, before even the fields approached here may be regarded as safely covered. And furthermore the gain is very modest if we only master a tiny corner out of the vast fields of test psychology, not to speak of other fields of psychology. I therefore feel an urgent need for investigations on a rather large scale and covering a wide range before we turn the methods thus acquired into, as it were, technological use.

The work presented in this book has grown out of practical tasks assigned to me as a consultant.

In 1945–1948 I analyzed data collected by the Department of Psychology at the University of Copenhagen with a view to a standardization of a new group intelligence test of the omnibus type, to be used by the Department of Defense in personnel selection. In carrying out the item analysis I became aware of the problem of defining the difficulty of an item independently of the population and the ability of an individual independently of which items he had actually solved.

In 1952 the Department of Social Welfare wanted a follow-up study carried out on a group of juveniles, 20 years old, which at school had attended special reading classes, because of reading retardation. This task resulted in the two models developed in chapters II and III, implying unambiguous definitions of ability and difficulty in connection with oral reading tests.

Towards the end of 1952 I became connected with the newly established Psychological Service Group of the Defense. A reanalysis of the omnibus intelligence test mentioned above was based upon the recently discovered

1. B. F. Skinner, *A Case History in Scientific Method*. The American Psychologist 11 (1956), p. 221–233.
2. J. Zubin et al., *Experimental Abnormal Psychology*. Columbia University Store. New York 1955. Mimeographed. – p. 2–28.

model of simple conformity (cf. chapter V and chapter VII, 12) and induced the construction of a new group intelligence test (BPP) consisting of four subtests, each designed to yield a measurement of a specific ability as introduced through the model. The analysis of subsequently collected data, demonstrated in chapter VI, shows how far the construction succeeded and where it failed.

The establishment in 1955 of the Danish Institute for Educational Research brought me a wealth of problems requiring clarifications, elaborations and extensions of the principles already laid down, the influence of which process pervades all the book.

In my attempts to provide for statistical tools pertaining to the psychological problems in question I have of course profited greatly from cooperation and from numerous discussions with many psychologists, too many to be named here.

The foundation, however, was laid in connection with my first mentioned work, through penetrative discussions with the late professor Edgar Rubin and his colleague E. Tranekjær Rasmussen, which I shall always recall with deep gratitude.

For the provision of conditions, necessary for coordinating my unorthodox research with the practical consultative work wanted by them I extend my sincere thanks to the institutions mentioned.

In particular I wish to express my gratitude to Major Poul Borking, M.A., head of the Psychological Service Group of the Defense in the period 1952-1956, and to Erik Thomsen, M.A., director of the Danish Institute for Educational Research, for the keenness with which they have at any time been ready to discuss my views, both theoretically and in their practical implications, and try to find means for carrying out my suggestions.

One of the consequences has been that in the final phase of my work I have had the privilege of drawing heavily upon the Institute. In particular I have utilized the facilities of the Department of Statistics with the leader of which, G. Leunbach, M.A., I have had a most fruitful cooperation since his appointment in 1956.

A preliminary Danish edition of the manuscript was carefully scrutinized by the staff members of the Institute.

The Danish text was translated – or rather transformed – into English by G. Leunbach, who has also revised later additions in English.

L. J. Savage of the University of Chicago has kindly read the final manuscript critically.

For all improvements to which the book has thus been subjected I am very thankful.

The publication of this work, nothing of which has previously been accessible outside a restricted circle, has been rendered possible by the generosity of the Institute. Thanks to that, this volume presents a lavishness of documentation – tables and graphs – which, although much needed, is nowadays not often seen in scientific publications.

The Danish Institute for Educational Research.

February 1960.

G. Rasch.

Part A

General principles and documentation

CHAPTER I

Introduction

1. Some general features of the present work

In the present work a new approach to test-psychology is attempted.

Traditionally the properties of a test are defined in terms of variations within some specified population. In practice such populations may be reasonably selected in various ways and accordingly the properties referred to, e.g. the reliability coefficient, are not specific to the test itself, they may vary, even strongly, with the population chosen.

In the following chapters we are going to deal mainly with three different types of tests and for each of them we shall develop a probabilistic model, in the application of which the rôle of the populations can be abolished.

As a consequence the statistical tools in current use in psychometric work, such as correlation coefficients, regression analysis, analysis of variance, factor analysis, etc. have found no place in our investigations. Even the normal distribution only plays the subordinate rôle of an approximation that may sometimes be helpful.

Each model implies two types of parameters, a "difficulty" for each test (or item) and an "ability" for each person. The "difficulties" of the tests of course have to be estimated from the body of data available, i.e. the results in two or more tests (or items) for each person of a certain collection. This collection, however, is not taken to be a sample from any "population". On the contrary, the estimation procedure may be so conducted that the personal parameters – the "abilities" – and their possible distribution are eliminated.

Similarly the "ability" of each person has to be estimated from the results of

the tests applied to him, but the estimation procedure yields a result that is independent of which particular set of tests (or items) has been employed.

Thus the parameters stand for intrinsic properties of the tests (or items) and the persons, and they can be estimated accordingly.

It follows that the concepts of measurement introduced through the definition of the two types of parameters differ radically from those employed in the psychometric theory of mental tests – we may, as an instance in point, just mention that in our theory we have no reason for considering a normal distribution of test scores as evidence for equality of units.

Our concepts are more akin to the psychophysical measurements in so far as these are concerned with individuals, each observed on several occasions.

The most conspicuous feature of our concepts, however, seems to be that they, in a certain well defined sense, carry the same conceptual status as mass and force in classical physics (cf. ch. VII, 7).

2. Desirability of eliminating populations

The desirability of being able to eliminate populations while evaluating the status of a particular individual was brought home to me in connection with the investigation of the career of reading-class pupils mentioned in the preface.

For the evaluation of the benefit these pupils may have drawn from this sort of education, various test results in mother tongue were available from the school period: before and after the transfer to reading classes, and either at the departure from school or on return to normal classes. The tests employed were oral reading tests, denoted ORF, ORU, ORS, OR₅ and OR₆, and two auditive spelling tests AS₁ and AS₂. To measure their level at the age of 20 years OR₆ and AS₂ were employed (cf. Appendix).

Now, of course, the tests used changed from one occasion to another, but none the less our aim was to evaluate the progress of each pupil. Thus it became an urgent problem whether it would be possible to ascertain the levels of attainment of a pupil independently of which tests were used and also independently of age, school group and time of school year.

If e.g. either oral reading test ORS or ORU can be used with a pupil, it ought to be possible to obtain essentially the same evaluation of his attainment no matter which test is used, i.e. the results of the two tests should be “translated” into the same “reading ability”.

Whether or not a set of attainment tests for the same area (e.g. oral reading tests ORF, ORU, ORS, OR₅ and OR₆) are equivalent in this sense cannot be decided a priori, that must be determined empirically.

3. Observational material

In order to do so, we had to collect a new material in such a way that tests could be compared directly two by two.

test	grade					
	2	3	4	5	6	7
ORF	+	+				
ORU		+		+		
ORS			+	+	+	
OR ₅			+	+	+	+
OR ₆					+	+

Table 1
Plan for collection of data for chaining
oral reading tests.

Table 1 shows the test program: Each oral reading test was given to children of all grades for which it had been used, and every child of grades 3 to 7 was given at least two tests. Besides, both spelling tests were given to all children.

From 24 schools four pupils of each of the grades 2 to 7 of primary school participated (and never more than two from one class). An even representation of sexes was obtained, and apart from this the children were selected at random. This material of 576 children forms the “normal” material, and it was supplemented by 144 pupils selected by teachers as poor readers and by a few very good readers.

4. On comparison of tests

As an example of the result of these experiments figs. 2 and 3 show the percentages of mistakes in spelling tests AS₂ plotted against those of AS₁ for pupils of the 2nd grade and 4th grade, respectively.

In accordance with the views set forth above we are not going to consider this as a case for correlation analysis. We shall quite unsophisticatedly glance at the figures and look for the possibility of translating uniformly from results in AS₁ to results in AS₂.

For the ordinary readers the percentage of mistakes usually keeps below 50, while it usually exceeds 50 for poor readers. But it is clear from the diagram that the two clusters of points lie in continuation of each other, both scattered around a straight line. The dispersion of the points about this line increases with larger values of the variables up to a certain point, but with percentages near 100% it again decreases. In the area covered by both groups of pupils they have the same dispersion.

Comparison of percentages of mistakes in oral reading tests OR5 and ORS, 4th grade.
 + good readers • representative sample • poor readers

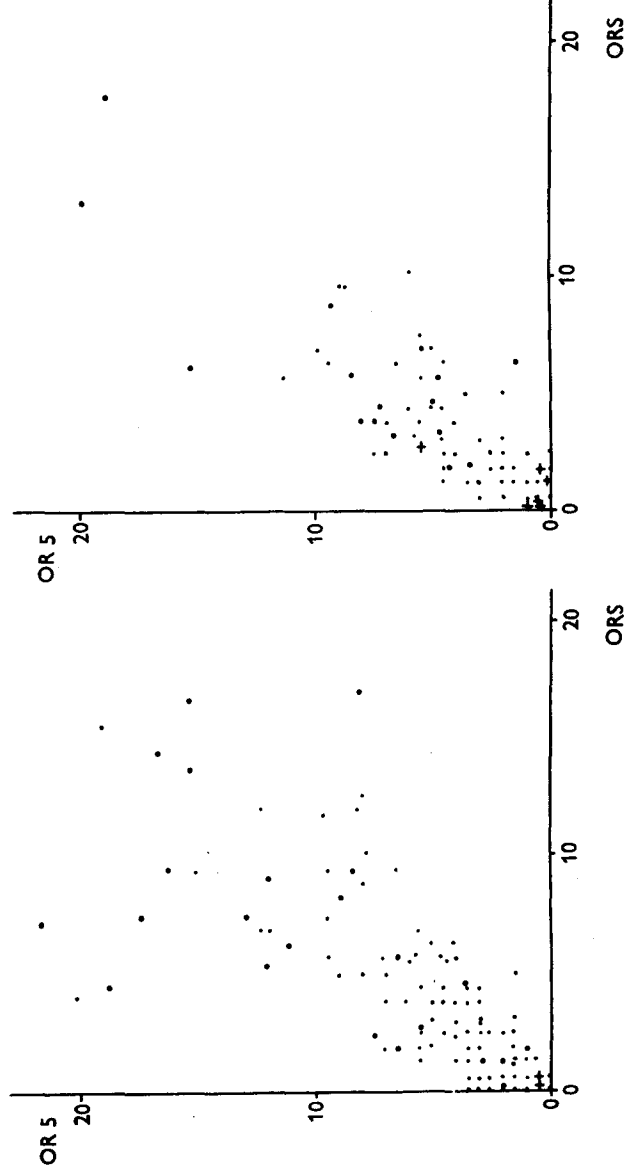


Figure 4

Comparison of percentages of mistakes in oral reading tests OR5 and ORS, 5th grade.
 + good readers • representative sample • poor readers

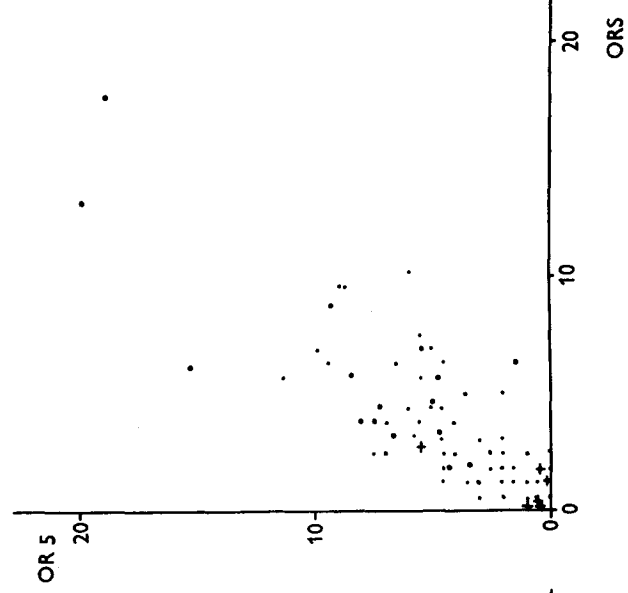


Figure 5

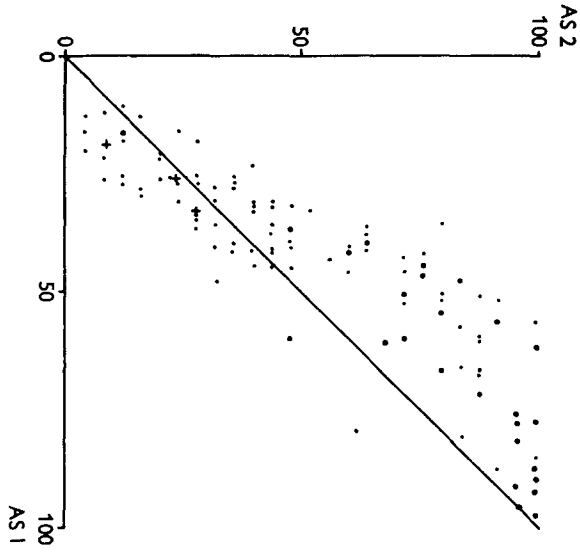


Figure 2

Percentages of mistakes in auditory spelling tests AS 1 and AS2, 2nd grade.

+ good readers • representative sample
 • poor readers

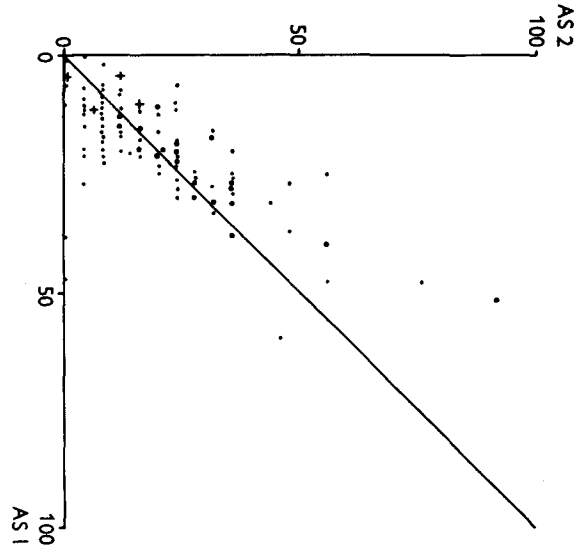


Figure 3

Percentages of mistakes in auditory spelling tests AS 1 and AS2, 4th grade.

+ good readers • representative sample
 • poor readers

Fig. 3 shows a similar picture, the two corresponding groups from 4th grade also continuing each other, both clustering around a common straight line. But furthermore the two straight lines almost coincide, i.e.: *the relationship between the percentage of mistakes in AS1 and AS2 is much the same, irrespective of grade and of whether normal pupils or poor readers are chosen.*

The same characteristic feature is found in several other materials, such as the relationship between percentages of misreadings in two oral reading tests (ORS and ORS₅) in the 4th and 5th grade (figs. 4 and 5).

For reading speed the same holds true. Figs. 6, 7 and 8 show the relationship between numbers of words read per 10 sec. in the same oral reading tests by pupils of grades 4, 5 and 6. Here the dispersion does not vary greatly with the test result, but the average number of words read varies very much from grade to grade and between the normal group and the poor readers. None the less the relationship between the two reading speeds is apparently the same in all groups.

These observations call our attention to a viewpoint which I have stressed in another context¹: *If a relationship between two or more statistical variables is to be considered really important, as more than an ad hoc description of a very limited set of data – if a more or less general interdependence may be considered in force – the relationship should be found in several sets of data which differ materially in some relevant respects.*

In relation to attainment tests all the school grades for which the tests are in practice applicable may be considered as forming a total collection of persons, that may be divided into subpopulations, such as single grades, sex groups and age groups within a grade, social strata, etc. Between the test results in such more or less extensive groups the same fundamental relationship may hold, and if so we shall use the term that *the relationship is "relatively independent of population"*, the qualification "relatively" pointing to the degree of breakdown that has been applied to the data.

In the cases mentioned above, as well as in many similar ones, a capital problem will be to describe those features of the relationship between the results of two or more tests which are independent of the populations.

In the last analysis the total collection or population may, of course, be broken down to single individuals, and that implies that the mathematical description of the relationship should comprise parameters that are specific to the individuals.

With the knowledge at our disposal to-day we may still in many cases have

1. Uppsala Symposium on Psychological Factor Analysis, March 17-19, 1953, Nordisk Psykologi Monograph No. 3, p. 68.

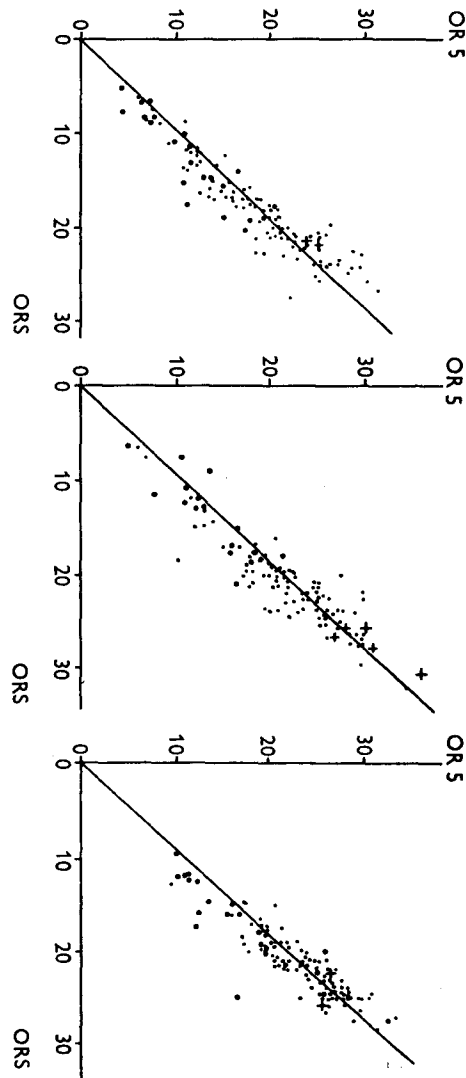


Figure 6

Comparison of reading speeds (number of words per 10 sec.) in oral reading tests ORS₅ and ORS, 4th grade.

Figure 7

Comparison of reading speeds (number of words per 10 sec.) in oral reading tests ORS₅ and ORS, 5th grade.

Figure 8

Comparison of reading speeds (number of words per 10 sec.) in oral reading tests ORS₅ and ORS, 6th grade.

• + good readers
• representative sample
• poor readers

• + good readers
• representative sample
• poor readers

• + good readers
• representative sample
• poor readers

to confine ourselves to ad hoc descriptions of those relationships. Such descriptions may be very valuable tools, but when a relationship has proved useful, the question why it is so easily arises. As long as we deal with populations or subpopulations an answer to such a question would seem beyond reach at present, to my knowledge no mechanism having been specified that may, within a population, produce the relations – including the variations – in the respects considered. But when we get down to individuals the task of deducing relations from basic assumptions seems much less hopeless. In fact, in this treatise I shall present some attempts to formulate and control such models for some attainment tests and some intelligence tests.

5. On models in classical physics

For the construction of the models referred to I shall take recourse to some points of view not especially connected with these tests, but of a more general character.

Into the system of classical physics enter a number of fundamental laws, e.g. the Newtonian laws.

A characteristic property of these laws is that they are *deterministic*. It is e.g. possible to calculate in advance the course which a pebble will follow if thrown under sufficiently specified initial conditions. In many fields this type of predictions can be made with an astonishing degree of precision, so close indeed, that aberrations between observations and calculated values have given rise to discovery of previously unknown factors, e.g. the presence of hitherto unknown planets.

None the less it should not be overlooked that the laws do not at all give an accurate picture of nature. They are very simplified descriptions of a very complicated reality. The laws describe an ideal universe, a model, on which reality may be mapped – leaving aside a lot of details. E.g. “a heavy point swinging without friction in a weightless string” never existed in the real world, but at a certain stage of the process of knowledge it is a very useful model of a pendulum.

This lesson seems worth keeping in mind when we attempt to build up models for what seems much more complicated, the behaviour of human beings.

In modern physics, beginning at the kinetic theory of gas, and ending up with atomic physics, the deterministic view has been abandoned. No deterministic description for e.g. radioactive emission seems within reach, but for the description of such irregularities the theory of probability has proved an extremely valuable tool.

6. On models in psychology

Where it is a question of human beings and their actions, it appears quite hopeless to construct models which will be useful for purposes of prediction in separate cases. On the contrary, what a human being actually does seems quite haphazard, none less than radioactive emission.

To return to our attainment tests: We may give a problem to a pupil of whom we know that he could easily solve it, and yet he fails. Or we may give him a task which is much too difficult, and anyhow he solves it. We can never know *with certainty* how a pupil will react to a problem, but we may say whether *he has a good or a poor chance of solving it*.

This way of speaking points to the possibility of mapping upon models of a kind different from those used in classical physics, more like models in modern physics – models which are *indeterministic*, where *chance* plays a decisive rôle: *The possible behaviour of a pupil is described by means of a probability that he solves the task.*

Since long ago probabilities have been known from games of chance. Consider a simple example.

In a ball-drawing game a ball is drawn at random from a bag with an equal number of white and red balls, the colour is recorded, the ball returned and the bag shuffled, and the game continues in the same way. Whether a single drawing will result in a white or a red ball is impossible to predict, but we say that *the probability of white is equal to 1/2*. This way of speaking implies among others that in a very long series of draws white and red will appear approximately equally often, so that the relative frequency of white will be very nearly 1/2.

If we consider an experiment consisting of 100 drawings, the result may well be 57 times red and 43 times white. If the experiment is repeated, white will generally appear from 35 to 65 times, most often in the neighbourhood of 50 times, in practice never 100 times or not at all.

If the experiment is repeated many times the number of white balls in each series can be ordered in a distribution, which will correspond very closely to that which can be computed by probability theory¹ – on the condition that the bag of balls is shuffled so thoroughly that successive drawings are independent of each other.

If a mouse is placed in a T-maze, it may turn right or left. If the mouse does not know the maze the probability of its turning right may reasonably be taken to be 1/2. If such an experiment is carried out with several mice, the

1. Various experiments of this type are reported in J. E. Kerrich, *An Experimental Introduction to the Theory of Probability*, Copenhagen 1946.

result may be put in the same terms as that of the ball-drawing experiment, on the condition that the behaviour of the various mice are "stochastically independent", i.e. that every mouse acts independently of the preceding mice, in the same way that each drawing was independent of the preceding ones.

The behaviour of the mice ought then to follow the *binomial distribution* which may be generally expressed thus¹:

We consider a process consisting of an unending series of steps. Each step results in either outcome A or $\bar{A}$, with probabilities θ and $1-\theta$, irrespective of the outcome of earlier steps (the outcomes are stochastically independent). The probability that a series of n steps results in the outcome A a times is then expressed by:

$$p \{ a | n \} = \binom{n}{a} \theta^a (1-\theta)^{n-a} \quad (6.1)$$

In the examples considered the steps are: each drawing, and the run of each mouse through the maze; the possible outcomes are: white or red, and run right or left; in both cases the probability $\theta = 1/2$, $1-\theta = 1/2$.

With e.g. 20 mice the probability of 10 runs to the right becomes

$$p \{ 10 | 20 \} = \binom{20}{10} \cdot \left(\frac{1}{2}\right)^{20} = 0.176. \quad (6.2)$$

By comparing experiments with such calculations, we may control whether the mice do behave in the way described by the model. In the words of the statistician: We may test the hypothesis that the model is as specified.

If we let the same mouse run several times, the model changes, the probability that it runs to the right is no longer $1/2$, if e.g. it learns that this is a dead alley. From a great number of observations we may estimate how the probability varies during such a learning process.

In the psychology of learning such models are becoming usual², but also in the evaluation of answers to single problems, e.g. the items of an intelligence test or an attainment test, it may be fruitful to speak of the probability of solving a problem correctly. The applicability of this point of view in the detailed analysis of tests will be discussed in chapter V.

1. See e.g. W. Feller, *An Introduction to Probability Theory and its Applications I*, 1. ed., New York and London 1950, ch. 6, 1-2.
2. See e.g. R. R. Bush and F. Mosteller: *Stochastic Models for Learning*, New York, 1955.

CHAPTER II

Oral reading tests. A model for misreadings

1. Poisson's law

In order to evaluate the number of misreadings in an oral reading test, we first consider the text as a series of words which, with regard to provocation of mistakes, are ordered very much at random. Further we make the bold hypothesis that a pupil with a certain degree of attainment has a certain probability of misreading any word of the text – for short: his *probability of misreading* – but that it is a matter of chance which words happen to be read wrongly¹.

If a pupil has a probability of misreading of 5% in a given text he may have 5 misreadings if the text contains 100 words, but the number may be somewhat larger or smaller. If he reads several texts of the same difficulty and length the numbers of misreadings will fluctuate around 5.

By the same reasoning as for the running mice, though with $\theta = 0.05$, we may compute the probability that he will make a given number a of mistakes by the binomial distribution:

$$p \{ a | 100 \} = \binom{100}{a} \cdot 0.05^a \cdot 0.95^{100-a}. \quad (1.1)$$

The probability of faultless reading becomes

$$p \{ 0 | 100 \} = \binom{100}{0} \cdot 0.05^0 \cdot 0.95^{100} = 0.95^{100} = 0.0059, \quad (1.2)$$

i.e. about 1/2%. The probability of one misreading is

1. For a considerable relaxation of this – rather severe – restriction, see ch. VII, 13.